Maths Teacher, 8A

Vasant Valley School, New Delhi

Subject: Maths

3139/M8A/0017

We are pleased to present the ASSET Teacher MyBook, a comprehensive resource designed to provide you with valuable insights into your students performance and support their academic growth. This resource offers a range of features to help you better understand your students' strengths and areas for development.

Personalized Teacher Feedback: Gain tailored insights into your students' distinctive strengths and areas for development. This feedback is aimed at assisting you in providing targeted guidance to each student.
Student Performance Table: Delve into the individual question-wise performances of your students. This data-driven view will aid you in identifying specific areas where students may require additional support.
Bird's Eye View: Acquire a holistic comparison of your students' performance against the national average. This perspective will help you contextualize your students' achievements and progress.
Year on Year Analysis - Previous Batch: This report compares student performance over a year.It compares the performance of a specific class compared to the previous year.
Skill-based Summary: Explore a tabulated display of students' skill proficiency, emphasizing the skills with the highest and lowest performance. This summary enables you to gauge skill performance relative to the school and national average.
Analysis of Easy/Difficult Questions: Identify questions posing ease or challenge compared to the national norm. This section illuminates areas needing focused attention, allowing you to tailor your teaching approach accordingly.
Questions with Common Wrong Answer: This section highlights potential misconceptions through selected questions. It provides valuable insights into your students' performance within a broader nationwide context.
Practice Questions and Explanation: In the Student MyBook, students receive practice questions tailored to their needs. This Teacher MyBook also features practice questions customized for your class, complete with answers and explanations. Different practice approaches have been thoughtfully curated based on subject and skill requirements.

We recommend that you review this analysis with reference to the ASSET question paper. Your commitment to your students' growth is highly commendable, and we believe that the ASSET Teacher MyBook will further empower you in this journey.

Should you have any questions or require further assistance, please feel free to reach out to us at info@ei.study. We are dedicated to supporting your efforts in enhancing your students' learning experience.

Regards,
The ASSET Team

Kindly note: Some images may slightly appear unclear due to technical reasons, but rest assured, this is not an error.

Dear Maths Teacher,

Congratulations on receiving your students' ASSET test results!

ASSET helps us understand where your students excel and where they might need more support.

Let's celebrate the strengths your students have shown in Maths are:

Data interpretation/analysis: concept of averages, graph reading, etc.
Applications in daily life: word/visual problems

Great job nurturing these talents!

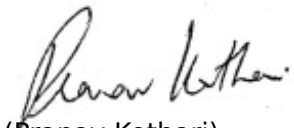
Now, let's look at areas where we can help them grow even more. Here's where they can improve in Maths :

Problem solving: advanced or challenging problems
Algebra: concepts and applications

To assist you in this journey, we've created a special booklet. It's designed to focus on the questions/ skills that need a little boost. The booklet provides insights into each student's progress and highlights questions that sparked interest or posed a challenge.

Take your time going through the analysis to understand how each student is doing. You'll find different sections in the booklet to spot learning gaps and dive into specific questions that need attention. Your students have personalized booklets, called Student MyBooks, too. These are tailored to their needs, giving them a glimpse into the areas where they can practice more.

Feel free to use these resources in class, and let us know how it goes. We're here to help and learn from your experience. Share your thoughts with us at info@ei.study.As you guide your students through their learning journey, you're helping them become even better. Best of luck!



(Pranav Kothari)
Chief Executive Officer - Educational Initiatives

In case of queries, please write to us at info@ei.study or contact our customer care executive at the toll free number 1800-102-8885 (9 am - 6 pm, Mon - Fri).

No.	Student	Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Score	Scaled Scores	Percentile	Performance
		Correct Answers →	C	D	A	B	A	C	C	D	C	A	A	B	D	A	C	C	D	D	C	D	C	B	C	D	D	B	C	D	D	C	C	B	B	D	C	C	D	C	D	B				
1	Aadvya Manhas		✓	✓	✓	✓	✓	B	✓	✓	D	B	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	A	D	A	C	B	✓	✓	✓	A	D	✓	✓	C	✓	✓	✓	B	B	✓	✓	26	618	87	
2	Aarav Chauhan		✓	C	B	✓	✓	D	✓	✓	D	C	✓	✓	C	✓	✓	✓	A	A	✓	A	✓	D	✓	C	A	A	B	B	✓	✓	✓	A	C	✓	A	✓	✓	✓	C	A	20	509	59	
3	Alisya Kalha		✓	✓	B	C	D	✓	A	✓	A	B	D	✓	C	B	D	✓	✓	✓	A	A	D	✓	B	C	A	C	B	B	A	D	✓	A	C	✓	B	A	✓	✓	B	A	13	436	25	
4	Amaira Shah		✓	✓	B	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	C	✓	A	D	C		✓	B	✓	✓	A	✓	D	A	C	✓	D	B	A	B	A	22	559	75	
5	Anshu Balmiki		✓	✓	C	✓	✓	✓	B	C	D	✓	C	✓	C	✓	✓	✓	C	✓	✓	✓	D	C	B	A	A	✓	✓	✓	✓	A	✓	C	✓	✓	✓	✓	C	B	C	✓	23	576	79	
6	Aryaveer Singh		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	C	D	37	800	99	O
7	Avni Garg		✓	✓	✓	✓	✓	A	✓	✓	D	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	✓	B	✓	C	✓	✓	✓	B	✓	✓	D	✓	✓	✓	C	A	✓	✓	31	705	96	
8	Gargi Malik		✓	✓	✓			B	A	C	D	D	✓	✓	✓	✓	B	B	A	✓	✓	✓	✓	D	✓	B	✓	C	✓	✓	✓		✓	D	✓	✓		✓	C		B	D	20	528	66	
9	Inaaya Irfan		✓	✓	✓	A	B	A	✓	✓	D	✓	✓	✓	C	✓	✓	✓	✓	A	D	C	✓	D	✓	B	✓	C	✓	✓	✓		✓	D	C	✓	✓	✓	C	B	✓	✓	24	564	76	
10	Jaiveer Bhutani		✓	✓	✓	✓	✓		✓	C	D	B	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	A	A	B	C	A	A	A	✓	✓	D	✓	A	C	C	B	A	B	✓	B	C	19	534	68	
11	Katha S Sahib		✓	C	C	A	✓	B	A	✓	B	C	✓	✓	A	✓	A	✓	B	✓	D	A	A	A	A	A	A	C	B	✓	A	B	✓	C	C	C	✓	✓	✓	✓	A	✓	15	473	44	
12	Kian Jaruhar		✓	A	✓	✓	✓	B	✓	✓	B	✓	✓	A	C	✓	✓	✓	B	A	✓	A	A	C	A	B	C	A	A	✓		D	B	✓	C	C	B	A	✓	B	B	C	15	459	36	
13	Kuvira Kumar		✓	✓	✓	A	*	A	✓	B	✓	B	✓	✓	C	D	✓	D	✓	✓	✓	✓	✓	D	B	C	B	A	✓	*	✓	D	✓	D	D	C	A	B	✓	A	✓	✓	19	494	53	
14	Kyra Wirk		✓	✓	✓	✓	✓	B	✓	✓	A	✓	✓	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	✓	✓	A	✓	✓	✓	B	C	D	B	D	30	643	91	
15	Mayank		✓	C	✓	✓	D	B	✓	A	✓	B	C	✓	C	B	✓	B	B	A	✓	A	A	✓	D	C	B	✓	A	B	✓	✓	✓	D	C	B	A	✓	B	✓	✓	A	16	454	34	
16	Nathania Chingsubam		✓	C	✓	✓	✓	B	D	✓	D	B	D	✓	C	✓	✓	✓	✓	✓	✓	✓	A	A	B	C	A	A	✓	✓	✓	D	✓	A	C	C	B	✓	B	✓	B	C	19	542	70	
17	Nikitha S Muthappa		✓	C	✓	✓	✓	B	D		✓	✓	✓	C	C	✓	B	✓	B	✓	✓	✓	✓	*	B	✓	A	A	✓	✓	✓	D	✓	D	✓	✓	✓	D	A	B	✓	✓	23	549	73	
18	Reanna Malik		✓	C	C	A	✓	B	A	✓	A	✓	✓	✓	B	✓	D	✓	B	✓	✓	C	A	A	A	A	A	A	B	B	✓	A	✓	C	C	C	✓	✓	A	✓	✓	✓	17	496	54	
19	Riddhhima Sahu		✓	A	✓	A	C	✓	B	✓	B	✓		✓	C	✓	✓	✓	✓	✓	✓	✓	A	A	B	C	A	D	B	B	A	D	✓	✓	D	C	✓	✓	B	A	✓	C	18	523	64	
20	Rudra Madan		✓	✓	C	✓	✓	B	✓	C	B	✓	D	✓	C	✓	B	✓	✓	✓	✓	B	A	C	B	✓	A	✓	B	B	✓	D	✓	A	✓	✓	✓	B	✓	✓	✓	✓	23	541	70	
21	Satya V Gupta		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	✓	✓	A	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	37	789	99	O
22	Shoaib Ahmad		A	✓	B	✓	✓	A	✓	✓	D	C	✓	✓	C	✓	✓	D	B	A	A	A	D	✓	A	C	A	D	✓	A	A	D	A	✓	C	✓	B	✓	B	A	C	C	14	436	25	
23	Shrey Gujral		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	39	800	99	O
24	Sifath Lohia		✓	C	✓	✓	✓	B	D	C	D	✓	D	✓	C	✓	B	✓	✓	B	✓	✓	A	A	A		B	A	D	B	A	D	✓	✓	A	✓	A	B	C	D	B	✓	15	489	51	
25	Soham Jha		✓	✓	✓	✓	✓	✓	✓	✓	B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	39	800	99	O
26	Sophie Singh		B	✓	C	✓	✓	✓	A	✓	B	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	B	B	A	C	A	✓	✓	A	✓	C	✓	✓	✓	✓	C	B	C	C	23	592	83	
27	Vani Garg		B	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	D	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	37	797	99	O
28	Vani Verma		✓	✓	✓	✓	C	A	✓	B	D	✓	✓	✓	C	✓	D	✓	C	A	D	A	✓	D	✓	C	A	C	A	C	✓	B	✓	A	A	C	A	✓	B	✓	C	A	16	477	46	
29	Veer Kumar		✓	B	D	✓	✓	✓	✓	✓	✓	✓	C	✓	C	B	D	✓	C	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	A	A	D	✓	A	✓	✓	✓	C	A	27	562	76	
30	Veer Ramchandani		✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	C	✓	✓	✓	✓	✓	B	✓	✓	38	800	99	O

% answered correct (section)	87	65	68	77	74	42	68	71	29	65	74	90	29	84	71	87	65	77	84	61	55	35	42	35	39	35	58	65	74	29	90	39	39	65	61	71	39	48	48	52	584.1/500			
% answered correct (school students)	78	58	63	75	71	38	66	69	32	58	76	73	38	80	48	86	57	71	69	61	53	31	34	34	22	44	49	60	58	28	89	36	50	65	56	70	44	59	53	57	561.1/500			
% answered correct (all students)	68	59	53	64	70	32	50	53	33	36	64	54	31	72	38	71	40	58	58	40	39	27	32	31	23	39	29	48	47	24	74	37	39	55	44	56	32	59	48	45	502.2/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. * Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

		Question No: →	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	Score	Scaled Scores	Percentile	Performance
No.	**Student↓**	**Correct Answers →**	**C**	**D**	**A**	**B**	**A**	**C**	**C**	**D**	**C**	**A**	**A**	**B**	**D**	**A**	**C**	**C**	**D**	**D**	**C**	**D**	**C**	**B**	**C**	**D**	**D**	**B**	**C**	**D**	**D**	**C**	**C**	**B**	**B**	**D**	**C**	**C**	**D**	**C**	**D**	**B**				
31	Viraaj Kathuria			C	✓	✓	D	✓	✓	✓	B	✓	✓	A	C	✓	✓	✓	✓	✓	✓	C	D	A	✓	✓	✓	C	✓	B	A	D	✓	✓	C	C	✓	✓	✓	✓	✓	✓	25	562	76	

% answered correct (section)	87	65	68	77	74	42	68	71	29	65	74	90	29	84	71	87	65	77	84	61	55	35	42	35	39	35	58	65	74	29	90	39	39	65	61	71	39	48	48	52	584.1/500			
% answered correct (school students)	78	58	63	75	71	38	66	69	32	58	76	73	38	80	48	86	57	71	69	61	53	31	34	34	22	44	49	60	58	28	89	36	50	65	56	70	44	59	53	57	561.1/500			
% answered correct (all students)	68	59	53	64	70	32	50	53	33	36	64	54	31	72	38	71	40	58	58	40	39	27	32	31	23	39	29	48	47	24	74	37	39	55	44	56	32	59	48	45	502.2/500			

Notes: Score: Actual score obtained by the student. **Percentile:** Percentage of students who have scored less than that particular student. **Performance:**'O' indicates outstanding performance in subject.
✓ Indicates correct answer. **A,B,C,D:** Indicates student's incorrect answer. **(Blank)** indicates student omitted answer. ***** Indicates student chose more than one option.
Scaled Score: Raw scores are converted to a scaled score through a process known as equating. The minimum a student can get is 200 and the maximum is 800.

The Bird's Eye View shows a comparison of the performance of your students with the national average performance.

Your class 8 A, has performed better than the national average in the Maths ASSET paper this year.

Class 8	Maths	No. of students
8A (Your Class)	117%	31
8B	111%	31
8C	109%	34
All sections combined	112%	96

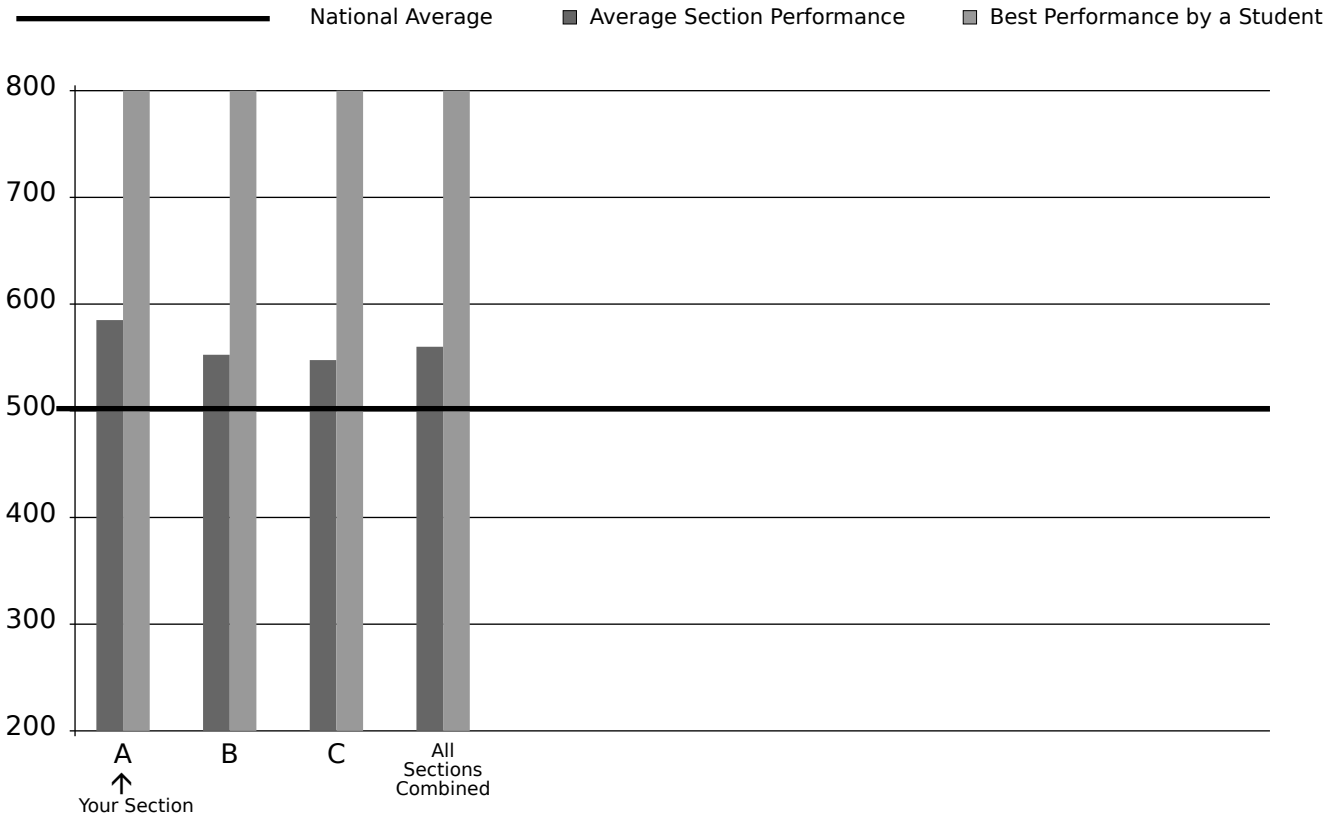
Performance as a percentage of National Average

Above Average(>105%) Below Average(<95%) Within Average(95% to 105%)

National average is taken as 100%. The value in the table indicate performance as compared to the national average.**E.g.** For a class average of 520 and overall average as 500, overall benchmarking %= (520/500)*100 = 104%. 95% to 105% is treated as similar to national average performance.Performance greater than 105% and below 95% is considered as performance above and below the national average range respectively.

Performance Data

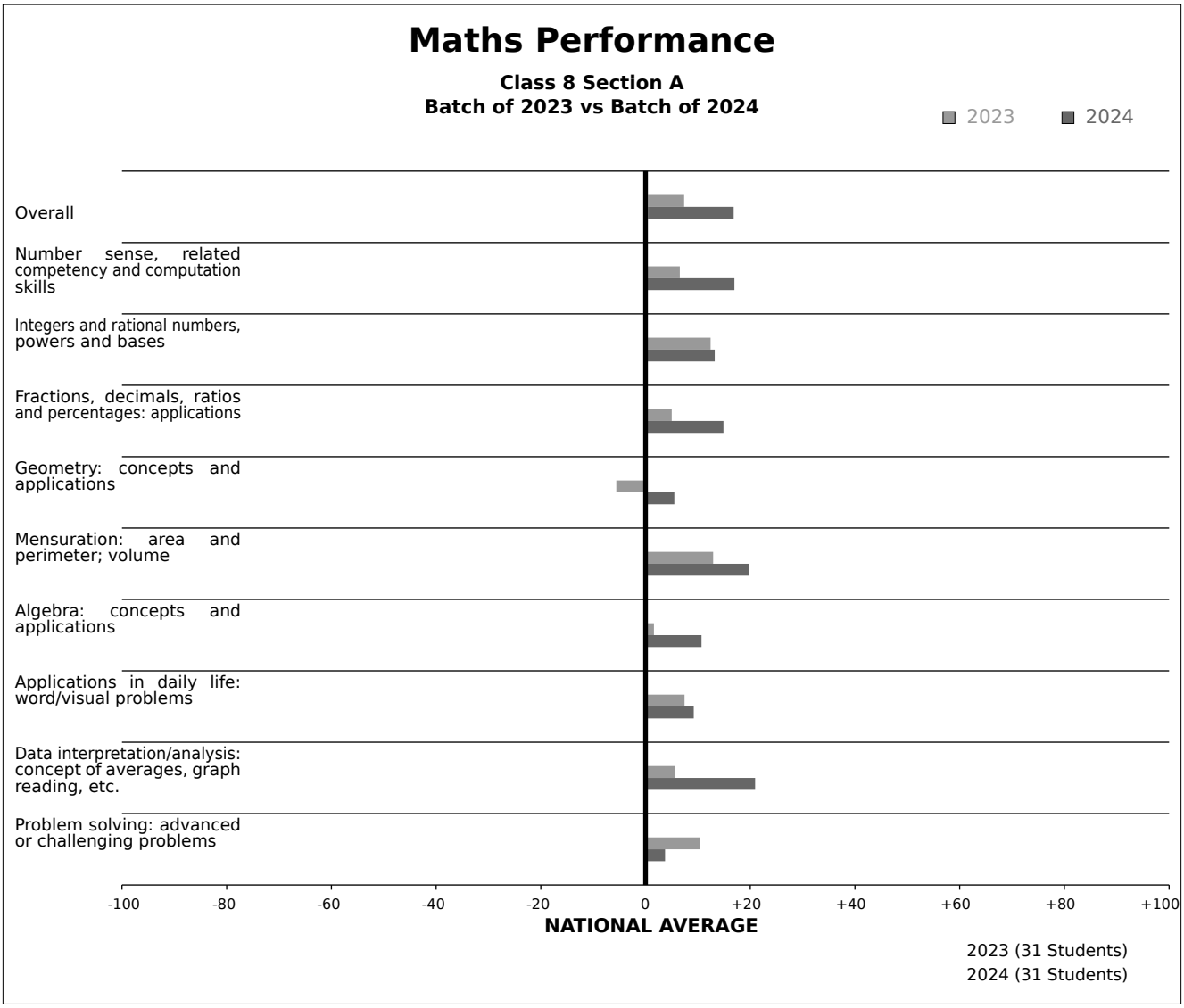
Comparison of section average and best performance with all schools in the country.



Comparison of performance of students of the current batch with the previous batch:

The following graph compares the performance of your current batch of students against the previous batch. The horizontal bars in the graph show how students of your class have fared against the national average in each skill across two years.

The thick black vertical line in the centre marks the national average.For example, a bar towards the right would mean that students performed higher than the national average that year in the particular skill.



This table shows how your students' performance compares to the overall school performance and the national average (all other schools in India). The skills with a '*' in the first column, if any, are those in which your students have performed better than the national average.

Class: 8 A Subject: Maths Number of Students: 31

S.No.	Skill	Questions	Class 8 A Performance	Class 8 Performance (School)	Class 8 Performance (National)
			Average	Average	Average
1*	Number sense, related competency and computation skills	3,8,19,23	66.1%	58.6%	49.2%
2*	Integers and rational numbers, powers and bases	13,14,16,20,35,36	65.6%	65.3%	52.4%
3*	Fractions, decimals, ratios and percentages: applications	11,18,21,25,29,39	61.3%	55.6%	46.4%
4*	Geometry: concepts and applications	1,9,28,38	57.3%	57.6%	51.8%
5*	Mensuration: area and perimeter; volume	4,10,15,24	62.1%	53.9%	42.3%
6*	Algebra: concepts and applications	6,17,22,33	45.2%	44.0%	34.5%
7*	Applications in daily life: word/visual problems	2,5,7,34	67.7%	64.8%	58.5%
8*	Data interpretation/analysis: concept of averages, graph reading, etc.	12,27,31,32	69.4%	61.7%	48.4%
9*	Problem solving: advanced or challenging problems	26,30,37,40	38.7%	43.2%	35.0%

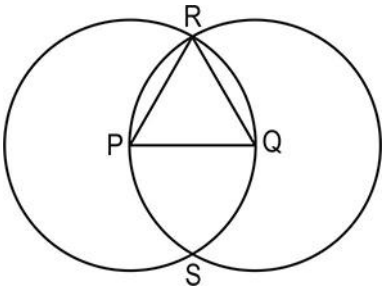
Note - The average performance is calculated from the number of correct responses given by students to the questions constituting specific skills.

Strong skills		Weak skills	
1.	Data interpretation/analysis: concept of averages, graph reading, etc.	1.	Problem solving: advanced or challenging problems
2.	Applications in daily life: word/visual problems	2.	Algebra: concepts and applications

This part of the report shows questions that the children of your class found easy and difficult. This information can be used by you to focus your attention to specific questions and skills. You can also use the student performance table to focus on individual students. (Please refer to the ASSET paper for the complete text of these questions.)

Questions which students of your school found the MOST DIFFICULT compared to the rest of the country:

Q 38 Two circles with centres P and Q are drawn such that each circle passes through the centre of the other. The circles intersect at points R and S as shown below.



(Note: The figure is not to scale.)

Which of the following is DEFINITELY true about ΔPQR?

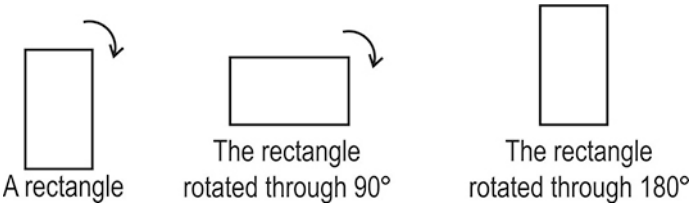
A It is a scalene triangle.
B It is an isosceles triangle.
C It is an equilateral triangle.
D It is a right-angled triangle.

Skill: Geometry: concepts and applications

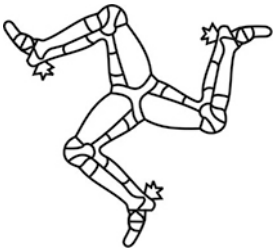
QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
38	28224	C	48%	59%	Your School	
					All Schools	

Q 9 Some shapes look exactly the same when they are rotated through a certain angle less than 360°.

For example, the rectangle shown below does not look the same when rotated through 90°. But it looks the same when rotated through 180°.



Shown here, is one such figure which appears on the flag of the Isle of Mann.



Through what angle must the above figure be rotated, so that it looks exactly the same?

A 30° **B** 60° **C** 120° **D** 180°

Skill: Geometry: concepts and applications

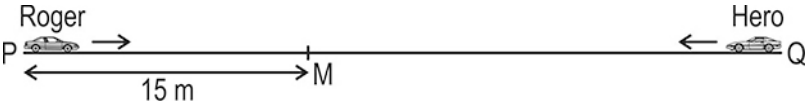
QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
9	28224	C	29%	33%	Your School	
					All Schools	

Q 26 Two toy cars 'Roger' and 'Hero' are placed at points P and Q which are 40 m apart. They start at the same time in opposite directions and meet each other at a point M which is 15 m from P.

After this -

- Roger continues towards Q at the same speed as earlier, and
- Hero reverses direction and goes back towards Q at the same speed as earlier. (Time taken to reverse direction is negligible.)

Which car will reach Q first?



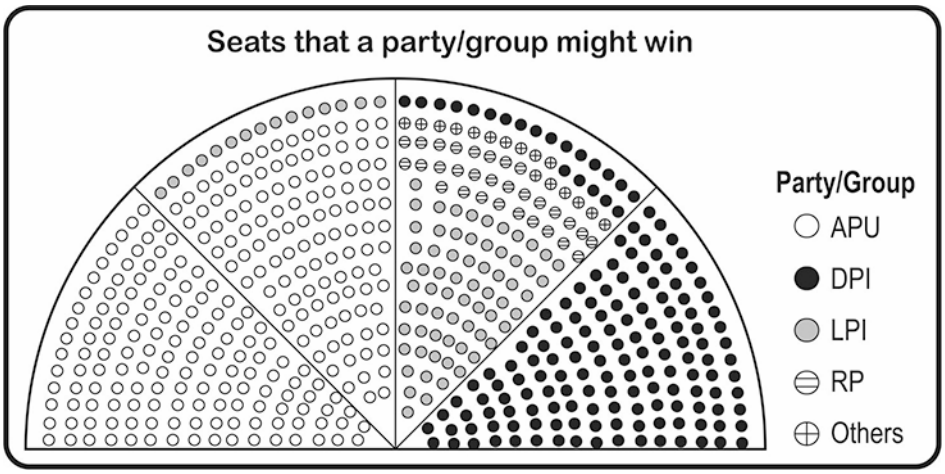
A Roger
B Hero
C (Both arrive at Q at the same time.)
D (We cannot answer without knowing their exact speeds.)

Skill: Problem solving: advanced or challenging problems

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
26	28224	B	35%	39%	Your School	
					All Schools	

Questions which students of your school found the EASIEST compared to the rest of the country:

Q 12 The figure shown here was found on a website showing the number of seats that different parties/groups might win in an election.



Roughly what percentage of the seats might the DPI win?

A 20% **B** 29% **C** 37% **D** 40%

Skill: Data interpretation/analysis: concept of averages, graph reading, etc.

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
12	28224	B	90%	54%	Your School	
					All Schools	

Q 15 The perimeter of the parallelogram PQRS shown below is 24 cm. PS = 5 cm, ST = 4 cm and ST ⊥ PQ.

(Note: The figure is not to scale.)

What is the AREA of the parallelogram?

A 14 cm^2 **B** 20 cm^2 **C** 28 cm^2 **D** 56 cm^2

Skill: Mensuration: area and perimeter; volume

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
15	28224	C	71%	38%	Your School	
					All Schools	

Q 27 The average temperature on the first three days of a week was 5°C.

If the average temperature on the first two days was 9°C, what was the temperature on the third day?

A 7°C **B** 3°C **C** -3°C **D** -7°C

Skill: Data interpretation/analysis: concept of averages, graph reading, etc.

QNo.	Paper Code	Correct Answer	% answered correct (your students)	% answered correct (all students)	Graph	
27	28224	C	58%	29%	Your School	
					All Schools	

From the questions that students across the country have found the most difficult, we have chosen a few to highlight potential misconceptions that students might have. For each question, we have given a break-up (in percentage form) of the number of students choosing a particular option. This has been done both for the section of your school and all the students nationally. Compare these to find out how your students have performed.

We have highlighted in light grey the correct option and in dark grey the most common wrong answer, and provided an explanation of why students are choosing the most common wrong option. It is recommended that you discuss these questions with the students to uncover students' rationale in choosing the most common wrong option. (Please refer to the ASSET paper for the complete text of these questions.)

Q 23 Which of the following is the BEST estimate of 8 ÷ 0.502?

A slightly less than 4
B slightly more than 4
C slightly less than 16
D slightly more than 16

This question tests whether students are able to estimate the quotient of a division problem involving decimals. Students were expected to realise that dividing 8 by approximately half gives an answer close to 16; then, as the divisor is slightly more than half, the quotient is slightly less than 16. Students who answered incorrectly either have the misconception that division always "makes smaller" or are not clear on the relation between divisor and quotient. Students can be helped to understand that "8 divided by half" means finding the number of "halves" in 8 - there are two halves in 1, so there are 16 halves in 8. Show them how this relates to the "invert and multiply" rule. Help students understand the relation between the parts of a division problem and the impact of increasing or decreasing the divisor. Beginning with whole numbers, guide them to realise that when we divide a fixed quantity into more number of parts, the size of each part will be smaller.

Papercode: 28224, Question no: 23					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	19%	29%	42%	10%	31
National	25%	26%	32%	15%	12084

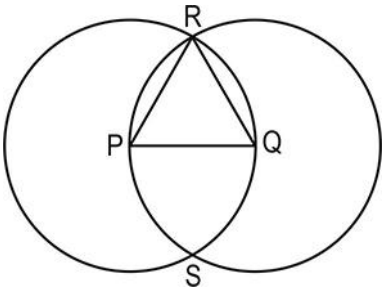
Q 22 p and t are two integers such that $(-p) \times (-t) < 0$.
Which of the following is true?

A both $p > 0$ and $t > 0$
B either $p < 0$ or $t < 0$
C both $p < 0$ and $t < 0$
D (Product of $(-p)$ and $(-t)$ can never be less than zero.)

The question tests understanding of integer multiplication. Students selecting option B have correctly understood that either (-p) or (-t) should be negative for the product to be negative. Students selecting option D thinks that (-p) and (-t) are always negative and hence the product can never be negative. Students selecting option C may have believed that as the expression in the question is less than 0, both p and t should be less than zero. Students selecting option A may have believed that multiplication of negative numbers is also negative. Explain that the expression (â^'p) represents the opposite of p, so if p is negative, (-p) becomes positive. Use examples where p is both positive and negative to illustrate this concept. Reinforce understanding by having students practice determining the sign of (â^'p) for various values of p.

Papercode: 28224, Question no: 22					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	26%	35%	10%	26%	31
National	12%	27%	23%	38%	12084

Q 38 Two circles with centres P and Q are drawn such that each circle passes through the centre of the other. The circles intersect at points R and S as shown below.



(Note: The figure is not to scale.)

Which of the following is DEFINITELY true about ΔPQR?

A It is a scalene triangle.
B It is an isosceles triangle.
C It is an equilateral triangle.
D It is a right-angled triangle.

This question tests the understanding of circle concepts. Option C is the correct answer as PQ = PR (radii of the first circle) and also PQ = QR (radii of second circle). Hence by transitive relation, triangle PQR is equilateral. Option A shows that properties of circles are not understood. Option B shows an error in understanding the given figure as they are unable to see that radii of both the circles are equal. Option D shows lack of geometry skills since resorted to visual estimation to answer the question.

Papercode: 28224, Question no: 38					
Performance	Option A	Option B	Option C	Option D	No. of Students
Section	16%	26%	48%	6%	31
National	10%	21%	59%	8%	12084

This section is crafted to aid students in practicing weaker skills. Use these in class discussions, gauge common understanding gaps, and tailor lessons. Answers and explanations are included. Employ a mix of question types to stimulate students' thinking. Here's how we've tailored the practice questions for each subject: For English, if reading comprehension is a challenge, we've provided full passages to work on. In Math, Science, and Social Studies, we've chosen questions based on areas where you've shown room for growth. Hindi practice focuses on non-reading comprehension skills. Computational Thinking covers a variety of skills across the class.

Skill: Problem solving: advanced or challenging problems (39%)

Q 1 A swimming pool has a length of K metres and a breadth of L metres. The 1 metre wide border of the pool (shown in grey) has to be tiled with 1 metre x 1 metre square tiles.

Which of the following formulae will give you the number of 1 metre x 1 metre tiles (T) required to make the border? (*Assume that K and L are whole numbers.*)

A T = K x L **B** T = 2(K+L) + 4 **C** T = (K x L) + 4 **D** T = K + L + 2

Explanation:

B: We require K tiles along the length and L tiles along the breadth. We have 2 borders of length K metres and 2 of length L metres. So to make these borders we will require 2(K + L) tiles. From the figure, are you able to see that this will leave the 4 corners uncovered? So we need 4 more tiles to complete the tiling, ie 2(K + L) + 4 tiles in all. K × L gives the number of tiles needed to cover the entire area enclosed by the swimming pool. That is not what we need to do. We just need to tile the shaded border shown in the figure.

Q 2 Look at the number chart given below.

1st Row →	1	2	3	4	5
2nd Row →	6	7	8	9	10
3rd Row →	11	12	13	14	15
...	16	17	18	19	20
...	21	22	23	24	25

The sum of the numbers in the first row is 15.
What will be the sum of the numbers in the 21st row?

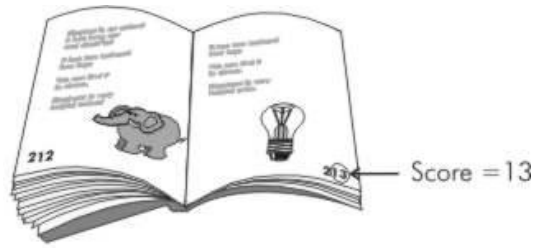
A 510 **B** 515 **C** 525 **D** 540

Explanation:

B: We see that there are 5 numbers in each row. So in 20 rows we would have 100 numbers. So the numbers in the 21st row would be 101, 102, 103, 104, and 105. Now the sum of these numbers is 515.

Each of the rows above has 5 consecutive numbers. Find the sum of the first few rows. Are you able to see a pattern emerging? Is it coming out to be 5 times the number in the middle? Can you find the sum 101 + 102 + 103 + 104 + 105 easily using this property?

Q 3 Raj and Priyanka are playing a game in which each person takes turns to snap open a book at a random page. The LAST TWO DIGITS OF THE PAGE NUMBER ON THE RIGHT SIDE is taken as the person's score.



A player's TOTAL score is calculated by adding each turn's score to the previous total. Which of the following could NOT be Priyanka's score after 4 rounds of the game?

A 6 **B** 88 **C** 145 **D** 250

Explanation:

C: Have you noticed the way the pages in a book are numbered? On which side do you see the odd numbers? Odd numbered pages are always on the right side of the open book right? So the score in each round will be an odd number. The score after 4 rounds would be the sum of 4 odd numbers, which as you can see, will always be an even number. So the score cannot be 145 which is odd. Scores of 1, 1, 1 and 3 in four turns are possible, and the total in this case is 6, which is a possible score.

Q 4 Numbers are to be filled in the following table in such a way that each row and each of the three 3×3 boxes contains EVERY DIGIT from 1 to 9.

	9			6			2	
				2		9		5
	4	2	?	8	3			6

Which of the following numbers should be filled in the shaded square?

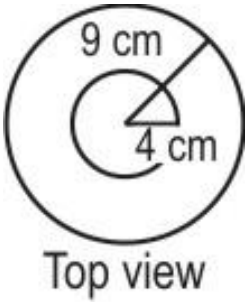
A 2 **B** 4 **C** 5 **D** 9

Explanation:

D: Each row and each of the 3 × 3 boxes must contain every digit from 1 to 9. There are only 9 small squares in each row and 3 × 3 box, so it means that no number can be repeated. See the position of 9 in the first 2 rows. There has to be a 9 in the third row, and it cannot be in the first or third blocks. So the 9 in the third row has to be in the middle block, and the shaded square is the only place where it can go. If we have 5 in the shaded square, the 9 of the middle block will have to be in the first or second rows which is not possible.

Q 5 Shown here is a tower with a circular base and a circular top. The slant height of the cone is 13 cm.

The radii of the top and the bottom circle are as shown in the figure below (top view).

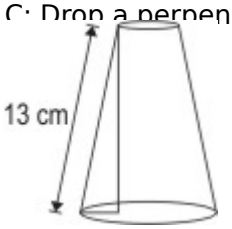


What will be the vertical height of the tower?

A 5 cm **B** 9.4 cm **C** 12 cm **D** 12.4 cm

Explanation:

C: Drop a perpendicular from the circumference of the circular top to the base.



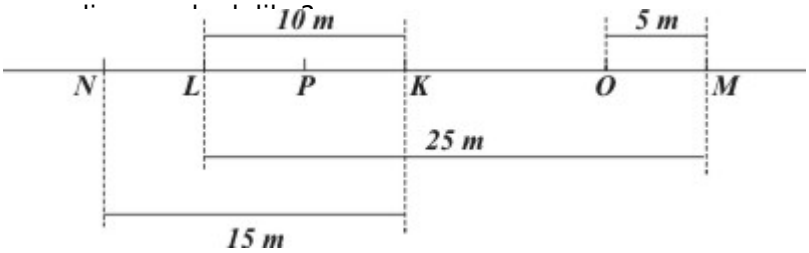
Are you able to see that the slant height of the tower, the vertical height and the difference in radii form a right triangle with the slant height as the hypotenuse? So applying Pythagorean theorem, we have $13^2 = 5^2 + (\text{vertical height})^2$. Solving we get $(\text{vertical height})^2 = 169 - 25 = 144$. So vertical height = 12 cm, as given in option C.

Q 6 Starting with the one who is first, what is their order at this point?

A L-P- N- K- O- M **B** N- L- K- P- M- O **C** L- N- P- O- K- M **D** N- L- P- K- O- M

Explanation:

D: Let us try to represent the given information by means of a diagram. K is 10 m behind L. L is 25 m ahead of M. So M would be 15m behind K is it not? M is 5 m behind O. So O would be between K and M and 5m from M. Mark the positions of the other people also in your diagram. What does



From the diagram can you not easily read off the order as N-L-P-K-O-M?

Q 7 Anand's printer has some typesetting error due to which exponents and multiplication signs are missed out.

For example,

2^5 X 9^3 = 23328 would appear in the print as

25 93 = 23328

He typed a correct equation but the equation that appeared in the print was -

$$210 \times 3100 = 3100$$

Which of these could have been the original equation?

A 2^{10} X 3^{100} = 3100
B 21^0 X 310^0 = 3100
C 21^0 X 3100 = 3100
D 21^0 + 3100 = 3100

Explanation:

C: We know that any number to the power 0 is 1. The number on the RHS of the equation in all the 4 given options is 3100. So we need to find what could be the exponents on the LHS so as to get 3100. Now 3^{100} is a very large number, and $2^{10} \times 3^{100}$ cannot be 3100. Also $21^0 \times 310^0 = 1$ and not 3100. Similarly $21^0 + 3100 = 3101$. Option C reads $21^0 \times 3100 = 1 \times 3100 = 3100$, thus C is the answer.

Q 8 Look at this number pattern.

$$1 + 3 = 4$$
$$1 + 3 + 5 = 9$$
$$1 + 3 + 5 + 7 = 16$$

How many consecutive odd numbers starting from 1, have to be added to get 64?

A five **B** six **C** seven **D** eight

Explanation:

D: From the given pattern, we see that the sum of the first 2 consecutive odd numbers is 2 × 2. The sum of the first 3 consecutive odd numbers is 3 × 3, the sum of the first 4 consecutive odd numbers is 4 × 4 and so on. 64 is 8 × 8. So following this pattern, we have to add the first 8 consecutive odd numbers to get 64.

Skill: Algebra: concepts and applications (45%)

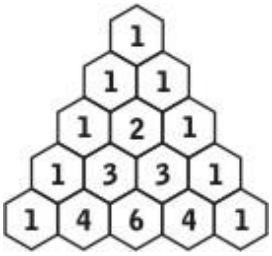
Q 1 If '2 n ' is the general form of writing any even integer, which of the following will NECESSARILY be an odd integer?

A n - 1 **B** 2 n - 1 **C** 2(n + 1) **D** 2 n + 2

Explanation:

B: 2 n is an even integer for any integer n . 2 n - 1 is one less than 2 n and hence will be an odd integer. What can you say about 2 n + 1? It is one more than 2 n and hence an odd integer, as well. But look at 2(n + 1). This is twice (n + 1), and therefore has 2 as a factor. So it is even. We may also say 2(n + 1) = 2 n + 2; ie, 2 more than 2 n , and hence its value is an even integer.

Q 2 See this number pattern carefully. Every row of numbers has a certain relation with the numbers in the row just above it.



Given below is a part of this pattern.

What number should replace **'*x*'** ?

A 7.5 **B** 5 **C** 4 **D** 3

Explanation:

B: See how the pattern is constructed. How would you write the next row? The first number is of course 1. The second number would be the sum of the first two numbers of the previous row, (1 + 4 = 5), the third number would be the sum of the second and third numbers of the previous row (4 + 6 = 10) and so on. Are you able to see that the number in each hexagon is the sum of the numbers in the two hexagons right above it? So in the given pattern, $x + 2x = 15$ ie $3x = 15$, $x = 5$.

Q 3 Ratnam hired a car from a car renting agency which charges a flat amount of Rs. x for travelling up to 40 kilometres and Rs. y per km for every kilometre travelled thereafter. How much should Ratnam pay the agency for a journey of 50 kilometres?

A $x + 10y$ **B** $50(x + y)$
C $40x + y$ **D** $50xy$

Explanation:

A: One has to pay Rs. *x* to travel up to 40 km and at the rate of Rs. *y* per km for every kilometre travelled above the first 40 km. Note that the rates are not uniform throughout the journey. The first 40 km are charged at a particular rate, and the remaining at the rate of Rs. y per kilometre. Ratnam travels 50 km. That is he travels 10 km above 40. He has to pay Rs. *x* for the first 40 km travelled. For the remaining 10 km he has to pay at the rate of Rs. *y* per km, which is Rs. 10 *y* . So totally he has to pay *x* + 10 *y* rupees.

Q 4 A new operation

★

is defined as follows:

p

★

q=2q - p for all rational numbers p and q.
According to this, what will be the value of 5

★

1?

A -8 **B** -3 **C** 8 **D** 9

Explanation:

B: Given that p★ q = 2q - p. So 5★ 1 = 2 × 1 - 5 = -3. Note that the rule is twice the second number (ie q) - the first number(ie p) So 5★ 1 = 2 × 1 - 5 and *not* 2 × 5 - 1 = 9.

Q 5 ♣ and ♦ are two numbers and ♣ + ♦ = 15.

The value of which of the following can be determined from this?

A ♣ − ♦ **B** ♦ × ♣

C 5♦ + 5♣ **D** 5♣ - 5♦

Explanation:

C:

The sum, ♣ + ♦ is 15. Now 5 ♦ + 5 ♣ = 5 ♣ + 5 ♦ = 5(♣ + ♦) = 5 × 15 = 75. 5 ♦ + 5 ♣ is nothing but 5 times the sum ♣ + ♦, and can be determined as we know this sum. Though we know the sum ♣ + ♦, we do not know the individual values of ♣ and ♦. The values could be 11 and 4, or 10 and 5 or even 14.5 and 0.5.Hence cannot find out the values of ♦ × ♣ or ♣ - ♦ from the given information alone.

Q 6 In a 2-digit number, *a* and b are the digits at its tens and ones place respectively.

Which of the following is the 2-digit number?

A $a \times b$ **B** $10\,a + b$
C $10\,b + a$ **D** $10\,(a + b)$

Explanation:

B: 74 is a two digit number with 7 at the tens place and 4 at the ones place. $74 = 10 \times 7 + 4$. We all know that 74 is not the same as 7×4. Similarly if we have a two digit number with a at the ten's place and b at the ones place, the value of the number would be $10 \times a + b = 10a + b$.

Q 7 The table below shows a relation between x and y:

x	1	2	3	4
y	1	5	9	13

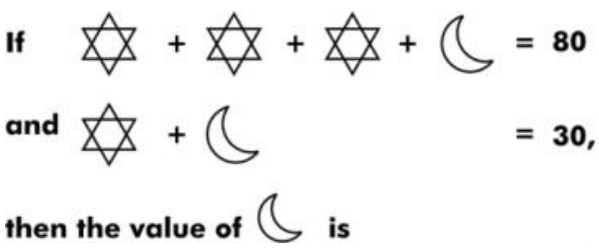
For what value of x will y equal 37?

A 8 ½ **B** 10 **C** 40 **D** 145

Explanation:

B: Study the table carefully. Are you able to figure out the relationship between x and y ? Remember that this should be satisfied by ALL pairs of x and y given. For example for the pair of values 3, 9, y is three times x . But this relation does not hold for all pairs of values. Here the y values are 3 less than four times the x values. ($y = 4x - 3$) So to find 4 times the x value, we add 3 to the y value. If the y value is 37, 4 times the x value is 40 (and not 37 - 3 = 34) Now the x value is $40 \div 4 = 10$.

Q 8

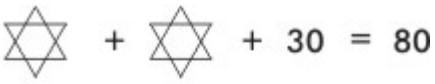


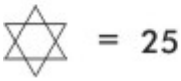
A 5 **B** 10 **C** 15 **D** 25

Explanation:

A: We are given that

. Substituting the value in the first equation and simplifying we get





So we see that star has the value 25. Using this in of the given equations we see that

= 5

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